



Determining the electron charge to mass ratio
using a Wehnelt-tube and Helmholtz-coil

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1 Introduction

This practical focuses on discovering the electric charge to mass ratio of the electron. One of the cornerstones of the current atomic model and aided in forming humanity's fundamental understanding of quantum physics, which lead to incredible feats of technological and computer engineering.

This experiment goes hand-in-hand with the discovery of the electron. Many individuals, throughout the centuries of the history of physics and philosophy, contributed to the discovery of this crucial elemental particle.

1.1 Historical background

Our understanding of atomic structure dates back to the ancient greek philosophers, such as Leucippus of Miletus (who is speculated to have originally thought of the then Greek philosophers understanding of atomic structure)[2]. John Dalton, alongside other noteworthy chemists and physicists such as Amedeo Avogadro and Joseph-Louis Gay-Lussac, formed the first understanding of atomic structures on a scientific level rather than a philosophical level. The model consisted of a few rules:

- All matter consists of small particles called "atoms".
- Atoms are indestructible and resist change.
- Elements are defined by their given masses.
- Atoms combined in whole number ratios that then lead to the development of "molecules".

This model, while flawed in some circumstances, laid incredible groundwork for future physicists and chemists to then further develop. One of the fundamental characteristics that his model does not contain is that of charged particles. Namely, protons and electrons.

For this report, the background shall mostly be focused on electrons as that is what this experiment is centered around.

J.J. Thomson was the first physicist to discover the electron. Charged particles were already discovered by the time J.J. Thomson began his work on the electron. Thomson's work proved that the Dalton model had its flaws as it did not contain any charged particles, and his work proved that atoms do contain charged particles that interact. Using this discovery, Thomson supported the "plum-pudding" model (created by Lord Kelvin)[3].

Although the means of this discovery will be delved into more deeply in the "method" section, the underlying principles are as follows: voltage is applied to a gas trapped in a tube. This leads to a flow of electric current through the gas. A light forms, as well as the emission of a ray of electrons (known as the

cathode ray). This cathode ray can then be manipulated with a magnetic field, proving that electrons are negatively charged particles[1].

The discovery of the atom, by J.J. Thomson, lead to several implications that challenged their current understanding of atomic models at the time. At the time, the hydrogen atom was considered to be the absolute smallest mass. However, this discovery challenged this notion as the mass was exceedingly small. Not only was the mass smaller, but electrons (or as he called it, corpuscles) seemed to be intrinsic to all matter. No matter what gas was used as a medium within the Cathode ray tube (or in this experiments case, the Wehnelt tube), the same corpuscles were emitted. It was also discovered that these particles were emitted by other mediums outside the experiment entirely, such as radioactive substances, ultra-violet light and other things. Thus leading to the conclusion that these corpuscles must be crucial to the design of the atom. As J.J. Thomson himself described it, "As these corpuscles are all negatively electrified, they will repel each other and so if an atom is a collection of corpuscles, there must in addition to the corpuscles be something to hold them together; if the corpuscles form the bricks of the structure, we require mortar to keep them together." [4]

1.2 Mathematical derivations applied

Note that this section is also largely the same as the mathematical derivations derived within the *Physics NPHY 22* Manual for Practica*.

To begin, we suppose that the electron has a charge denoted by e . The force a charged particle experiences in a magnetic field $\vec{B}$ is expressed by:

$$\vec{F} = e(\vec{v} \times \vec{B}) \quad (1)$$

if we assume this is the net force the charged particle experiences, then we can apply Newton's 2nd law as follows:

$$m \frac{d\vec{v}}{dt} = e(\vec{v} \times \vec{B}) \quad (2)$$

we then scalar multiply both sides by the magnetic field $\vec{B}$ to yield

$$\vec{B} \cdot \frac{d\vec{v}}{dt} = e\vec{B} \cdot (\vec{v} \times \vec{B}) = 0 \quad (3)$$

the reason the right hand side is 0 is that the dot product between a vector and 2 vectors being cross product, where one of the vectors is the dotted vector, always yields 0. We also assume $\vec{B}$ to be a constant value such that $\frac{d\vec{B} \cdot \vec{v}}{dt} = 0$ and thus $Bv \cos \theta = K$, where K is some constant value. Because the values of B and V remain constant, we can thus assume that the angle θ must also remain the same. This proves the first part of the experiment, where angle is compared to radius. The next step of the derivation is ensuring the constancy of the radius of the electrons.

The net force the electrons experienced can also be expressed as:

$$e(\vec{v} \times \vec{B})Bv\sin\theta = \frac{mv^2}{\rho} \quad (4)$$

where ρ is the radius of the trajectory the electron beam follows. This then simplifies to:

$$\rho = [\frac{v\sin\theta}{B}]/\frac{e}{m} \quad (5)$$

Because v, B and θ all remain constant, radius will then also remain constant. When we assume that $\theta = 90^\circ$. The relationship between the radius of the trajectory and the radius of the cylinder ρ_0 is:

$$\rho\sin^2\theta = \rho_0 \quad (6)$$

We factor ρ_0 rather than the trajectory radius as that is the radius that will be measured in the experiment. Thus the equation, with the new radius substituted in, becomes:

$$\rho_0 = [\frac{v\sin\theta}{B}]/\frac{e}{m} \quad (7)$$

With the velocity, v , being in the equation, one can factor in the equation for kinetic energy which can be expressed as:

$$eV = \frac{1}{2}mv^2 \quad (8)$$

thus the final equation used will then become:

$$\rho_0 = \frac{\sqrt{2V}(\frac{\sin\theta}{B})}{\sqrt{\frac{e}{m}}} \quad (9)$$

Now another key mathematical formula needed is that of the magnetic field created by the Helmholtz coils. To start, we use the equation of a magnetic field of a flat coil at a distance y from the centre of the coil.

$$B = \frac{1}{2}\mu_0 N i a^2 (a^2 + y^2)^{-1.5} \quad (10)$$

where i is the current, and N is the number of windings within the coil. Suppose the 2 Helmholtz coils are placed a distance $2b$ from one another, and suppose their currents flowed in the same direction, which makes their respective magnetic fields point in the same directions as well. The magnetic field at a point can then be denoted as:

$$B = \frac{1}{2}\mu_0 N i a^2 [(a^2 + (b+x)^2)^{-1.5} + (a^2 + (b-x)^2)^{-1.5}] \quad (11)$$

This equation, when expanded, will then become a series, where the first term yields the magnetic field at the centre of the distance between the coils, and

the other terms give the magnetic field strength at points that deviate from the centre. Allowing $2b = a$ will make the 2nd term disappear, which will then simplify the formula further.

$$B = \mu_0 N i a^2 \left(a^2 + \frac{a^2}{4}\right)^{-1.5} = \frac{4^{\frac{3}{2}}}{5} \frac{\mu_0 N i}{a} = (7.8 \times 10^{-4}) i \quad (12)$$

2 Experimental set up

2.1 Apparatus

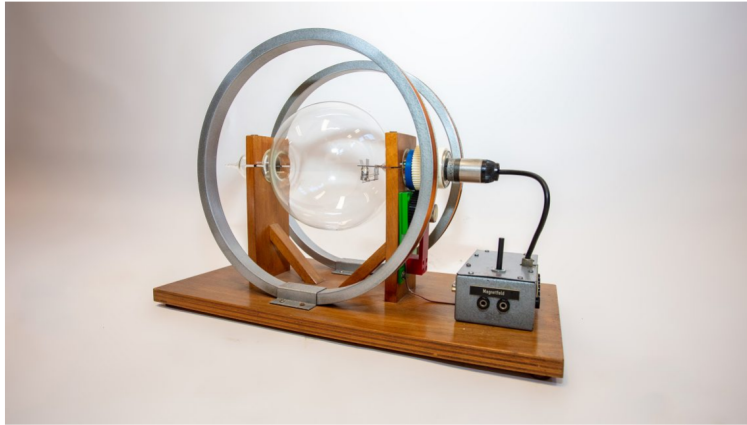
The following apparatus were used within this experiment:

- Wehnelt Tube
- 2 Helmholtz coils
- A voltage supply for both the Wehnelt tube and 2 Helmholtz coils

2.2 Method

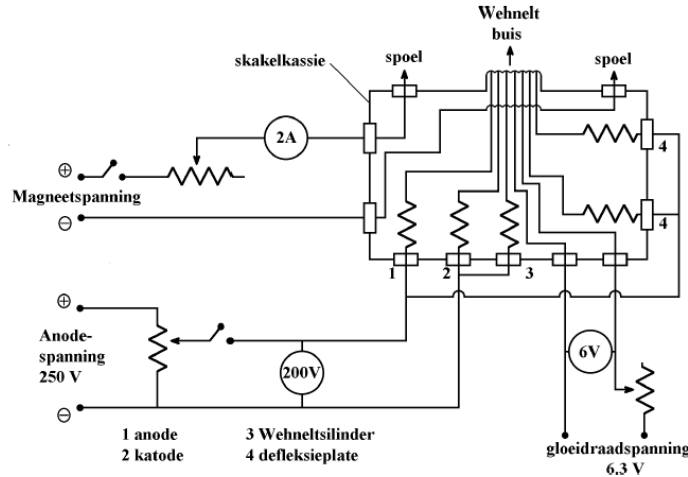
Do note that the following subsection is based largely off of what is documented within the *Physics NPHY 22* Manual for Practica* as it provides a thorough description of the process being used in this experiment.

Set up



Source: <https://tecnico.ulisboa.pt/en/events/tecnico-podcast-110-historias-110-objetos-the-wehnelt-cylinder/>

1. The Wehnelt tube and Helmholtz coil is both connected to their respective voltage supplies as according to the following circuit diagram:



Source: *Physics NPHY 22* Manual for Practica, 2024*

2. Apply approximately 6.3V to the filament connected to the cathode.
3. Once the connection has been successfully made, allow the filament within the Wehnelt tube to heat up. This process should take about 2 minutes.
4. Switch the voltage connected to the anode on. Note that as anode voltage is increased, the intensity of the light emitted by the electron beam is increased.
5. Slowly increase the applied voltage to the anode so that the electron beam becomes more visible. Note that higher voltages widens the beam. The beam should be forming a helical shape.
6. Decrease the voltage of the filament in order to maintain the visibility of the electron beam.

This concludes the set up for both parts of the experiment. The first part focuses on the dependence of the angle of the electron beam compared to the magnetic field induced by the Helmholtz coil.

Relationship between θ and ρ_0 For this experiment, we measure the angle θ between the Wehnelt tube and Helmholtz coil. This method will prove one way of determining the electron mass to charge ratio.

1. We set the anode at a voltage $\pm 120V$
2. Ensure the Wehnelt tube is first aligned at 90° .
3. Then, rotate the Wehnelt tube to 80° and measure the corresponding radius ρ_0 .
4. Repeat this process until you reach 50° . Document results.

Relationship between ρ_0 and V and B The tube is now returned back to $\theta = 90^\circ$. Then follow the following process for determining the relationship between ρ_0 and voltage and magnetic field is followed:

1. Set the voltage through the anode to 150V.
2. Change the current that passes through the coils then change the magnetic field.
3. Measure the change in radius ρ_0 caused by the change in magnetic field.
4. Do multiple measurements of ρ_0 with varying current values.
5. Repeat this process with voltage values of 200V and 250V.

3 Results

The following section consists only of the data collected and graphs based upon the collected data. The first section discusses the mathematical formula used to determine the electron mass to charge ratio, the 1st part is using the data gathered from the first section of the experiment, and the 2nd part is about the data gathered from the 2nd part.

Derivation for mass to charge ratio Based on equation (9), we can get the ratio as follows:

$$\rho_0 = \frac{\sqrt{2V}(\frac{\sin\theta}{B})}{\sqrt{\frac{e}{m}}} \quad (13)$$

Dividing both sides by ρ_0 and then multiplying both sides by $\sqrt{\frac{e}{m}}$, then squaring both sides of the equation will yield:

$$\frac{e}{m} = \frac{2V(\frac{\sin\theta}{B})^2}{\rho_0^2} \quad (14)$$

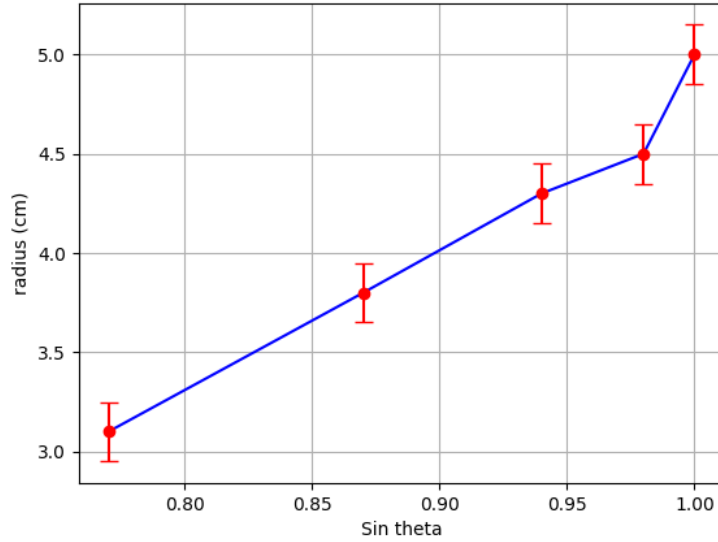
This is the equation that will be used to derive the ratio.

3.1 Figures: Experiment part 1

This section focuses entirely on the first part of the experiment. The current that was flowing through the Helmholtz coils is 1.55A. Thus, according to equation 12 ($7.8 \times 10^{-4}i$), the magnetic field is $1.209 \times 10^{-3}T$

Part 1				
Angle(θ)	sin(θ)	Radius(m)	e/m	Standard deviation
90	1	5,00E-02	8,76E+10	1,74E+10
80	0,98	4,50E-02	1,04E+11	
70	0,94	4,30E-02	1,05E+11	
60	0,87	3,80E-02	1,15E+11	
50	0,77	3,10E-02	1,35E+11	

Table 1: Angle measurements compared to the radius of electron beam
Average of calculated e/m values: $1.0932 \times 10^{11}C/kg$

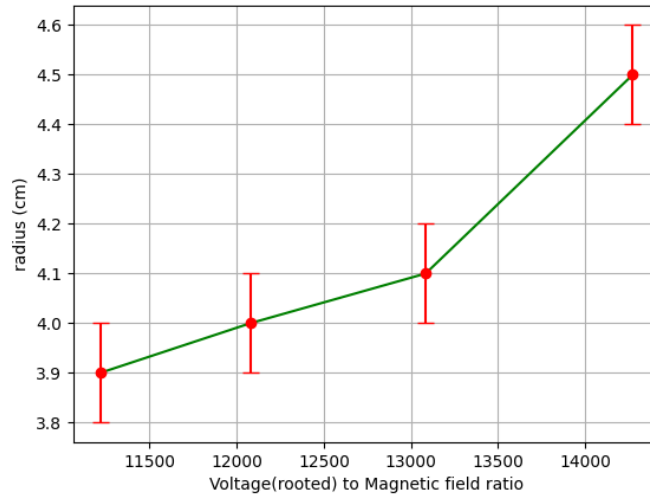


Graph 1: $\sin\theta$ compared to the corresponding radius, with 0.15 error margins

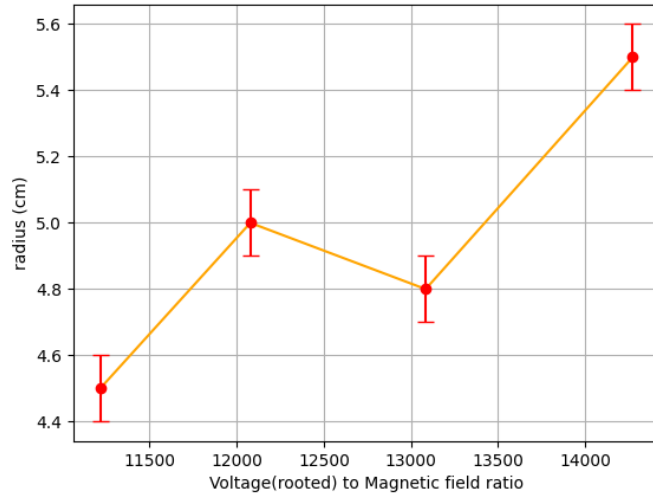
3.2 Figures: Experiment part 2

V=150				
Current(A)	Magnetic Field (T)	Radius(m)	rooted Voltage over Magnetic field($V^{(0,5)}/T$)	e/m value
1,1	8,58E-04	0,045	1,43E+04	2,01E+11
1,2	9,36E-04	0,041	1,31E+04	2,04E+11
1,3	1,01E-03	0,04	1,21E+04	1,82E+11
1,4	1,09E-03	0,038	1,12E+04	1,74E+11
Standard Deviation:	1,25E+10			
V = 200				
Current(A)	Magnetic Field (T)	Radius(m)	rooted Voltage over Magnetic field($V^{(0,5)}/T$)	e/m value
1,1	8,58E-04	5,50E-02	1,65E+04	1,80E+11
1,2	9,36E-04	4,80E-02	1,51E+04	1,98E+11
1,3	1,01E-03	5,00E-02	1,39E+04	1,56E+11
1,4	1,09E-03	4,50E-02	1,30E+04	1,66E+11
Standard Deviation:	1,60E+10			
V = 250				
Current(A)	Magnetic Field (T)	Radius(m)	rooted Voltage over Magnetic field($V^{(0,5)}/T$)	e/m value
1,1	8,58E-04	6,00E-02	1,84E+04	1,89E+11
1,2	9,36E-04	5,90E-02	1,69E+04	1,64E+11
1,3	1,01E-03	5,00E-02	1,56E+04	1,95E+11
1,4	1,09E-03	4,70E-02	1,45E+04	1,90E+11
Standard Deviation:	1,19E+10			

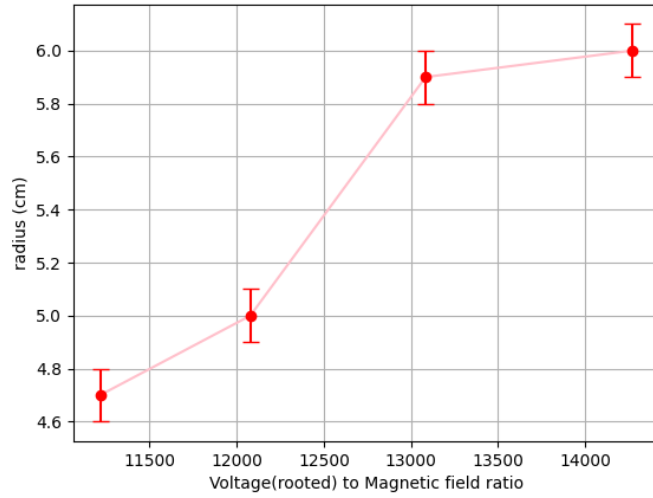
Table 2.1 - 2.3, Voltages and the currents passing through the Helmholtz coils, along with e/m values.



Graph 1: Voltage = 150V, $\sqrt{V}/B$ vs ρ_0



Graph 2: Voltage = 200V, $\sqrt{V}/B$ vs ρ_0



Graph 3: Voltage = 250V, $\sqrt{V}/B$ vs ρ_0

4 Discussion

This discussion, similarly to the results, will be divided into two sections. Section 1 pertaining to the first part of the results, and section 2 pertaining to the 2nd part of the results.

4.1 Results part 1: discussion

Upon looking at the graph and the table, the relationship between angle and radius ρ_0 clearly exhibits a linear relationship within the given error margin (0.15), although with a noticeable outlier at the 4th measurement taken. This could be attributed to error in reading the device properly by the participants.

When examining the calculated e/m values, the average came to 1.0932×10^{11} C/kg. Although there was a strange upward trend of the e/m values, when it should be a "static" value. This could again be attributed to either incorrect measurement techniques or human faults in the procedure of the experiment.

4.2 Results part 2: discussion

When examining the figures of each of the experiment trials in each set of voltages, large inconsistencies with mathematical models and measured data appear. The mathematical models clearly exhibit linear relationships between each of the variables when ρ_0 is the subject of the equation. Graph 1, largely, maintains the linear relationship predicted by the model, but graph 2 and 3 show drastic outliers within the data. Similarly to the first section, this may be attributed to error in measuring the data itself. This will be discussed in the next section.

When examining each of the tables e/m sections, the data does seem to not vary as much as the previous data did. The first set has a significant larger standard deviation when compared to standard deviations calculated by from other tables. Within this section of the experiment, when voltage was at 200V, the standard deviation was at its greatest. Then when the voltage was 250V the standard deviation was at its minimum.

These outliers can possibly be attributed to the process of measuring the radius. To measure the radius, one had to see past the beam of electrons and compare it to the ruler behind to measure. However, this proved (at points) difficult to accurately measure as the beam of the light had subtle movements and the bright light outshines the ruler, so it was difficult to see the values upon the ruler. There was enough data to give a good indication of where the electron charge to mass ratio may lie.

5 Conclusion

The aim of this experiment was to find the electron charge to mass ratio using a Wehnelt tube and Helmholtz coil, as the negative charge of the electron will be effected by the magnetic field generated by the coil. While the experiment failed to yield a precise result (result with minimal standard deviation), it still gave a very clear indication that the electron charge to mass ratio must lie within some value that is $\pm 1.5 \times 10^{11}$ C/kg. What this experiment demonstrates clearly is the potential of this set up if further upgrades were

applied. The means of measuring the radius proved difficult and may have cause a sum of inaccuracies within the measurements, leading to several outliers in the data set. The implication of this discovery is truly ground breaking as well. The implications in defining an accurate atomic model expanded far more as a result of this discovery. Due to the presence of the negatively charged electron, atoms could then be manipulated and study using magnetic fields, which could then lead to even further research in the quantum fields and now has far more application in fields such as CPU (central processing unit) design and other computer related design.

6 Bibliography

References

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- [2] Ken Stewart. *Atomic Model*. Nov. 17, 2023. URL: <https://www.britannica.com/science/atomic-model>.
- [3] George Paget Thomson. *J.J. Thomson*. June 7, 2024. URL: <https://www.britannica.com/biography/J-J-Thomson>.
- [4] Joseph John Thomson. *The structure of the atom*. Academie Royale de Belgique, 1913.

sources

7 Questions

This section focuses on the questions asked within the practical guide.

1. Constant magnetic field: One that does not change over time.
 Uniform magnetic field: A magnetic field that has the same value at any point along its field.
 Homogeneous magnetic field: The force of the magnetic field is constant at all points of interaction.
2. Electrons can be accelerated using cathode rays at appropriate voltage levels, which was often used in older models of televisions.
3. Yes. It is the smallest charge and is also known as the "elementary charge" at a value of 1.6×10^{-19} .
 Source: <https://www.khanacademy.org/science/up-class-12-physics/x0958a876c1afdc76:electric-charged-and-fields/x0958a876c1afdc76:electric-charge/a/electric-charge-ap1>

4. Highly energies ions (due to high voltages) collide with atoms in the atmosphere. The collision then excites the atoms, which in turn makes said atoms return to ground states that then emit a photon at a wavelength of blue light (450nm to 495nm).
5. Helmholtz coils are typically used in development of navigation systems, as well as the development of magnetometers.
Source: <https://www.pnisor.com/helmholtz-coil-design-applications-and-impact/>
6. It is the 4th fundamental state of matter where electrons can move freely from the nuclei they were bonded to.
Source: <https://www.techtarget.com/whatis/definition/plasma>
7. Yes. In cancer treatment there is a therapy called "Proton beam therapy" which fires a beam of protons towards a tumor.
Source: <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/proton-therapy>
8. Yes. Because of the elementary charge the electron contains, and the linked nature of electric fields and magnetic fields, the electron can be moved by magnetic fields.
9. The magnetic poles of the earth pull and push charged particles around the atmosphere of the planet.
10. An aurora is a phenomena that occurs in the earths atmosphere when it interacts with solar winds. The charged particles from the solar winds move due to the magnetic poles of the earth. *Source: <https://www.scirp.org/journal/paperinformation?paperid=123131>*